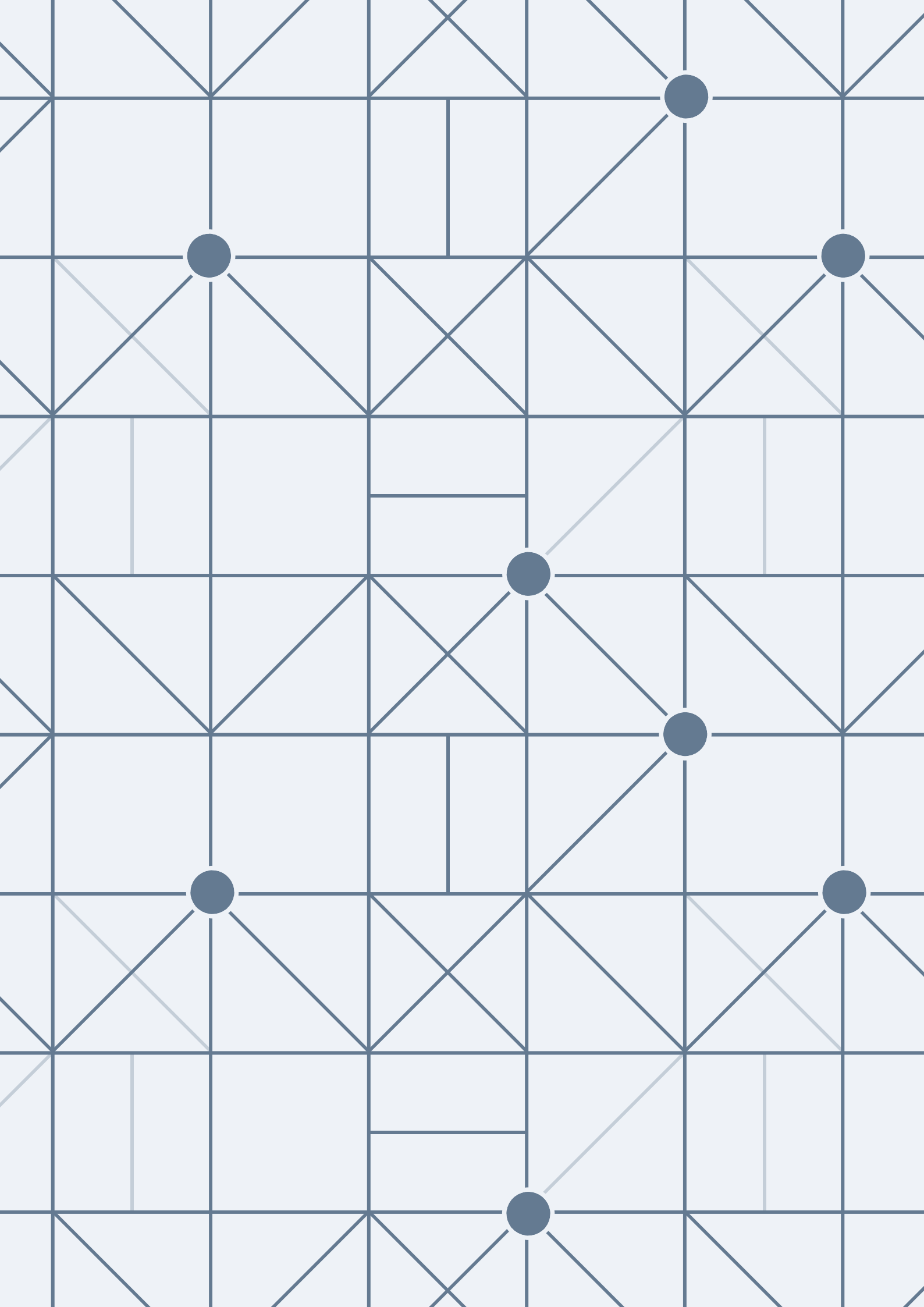
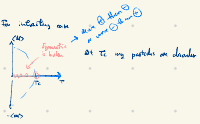




$H = -\frac{1}{2} \sum_{i,j} \frac{q_i q_j}{r_{ij}} \Rightarrow$ Act interacting many body
 where $q_i = e, -e$
 $r_{ij} = |r_i - r_j|$

or
 At
 $U = U(r)$?
 $U(r) = \frac{1}{2} \sum_{i,j} \frac{q_i q_j}{r_{ij}}$?



$z = (2\pi)^{-3}$
 $z = \frac{1}{(2\pi)^3} \int d^3p e^{i\mathbf{p} \cdot \mathbf{r}} = \delta^3(\mathbf{r})$

$z = (2\pi\hbar)^{-3}$
 $z = \frac{1}{(2\pi\hbar)^3} \int d^3p e^{i\mathbf{p} \cdot \mathbf{r}} = \delta^3(\mathbf{r})$

$U = \left(-\frac{1}{2} \sum_{i,j} \frac{q_i q_j}{r_{ij}} \right) = \frac{1}{2} \sum_{i,j} \int d^3r d^3r' \frac{q_i q_j}{|\mathbf{r} - \mathbf{r}'|} \delta^3(\mathbf{r} - \mathbf{r}_i) \delta^3(\mathbf{r}' - \mathbf{r}_j)$
 is this way $U = U(r) = -\frac{1}{2} \sum_{i,j} \frac{q_i q_j}{r_{ij}}$
 $U(r) = -\frac{1}{2} \sum_{i,j} \frac{q_i q_j}{r_{ij}} = -\frac{1}{2} \sum_{i,j} \frac{q_i q_j}{r_{ij}}$
 where $U(r) = \frac{1}{(2\pi\hbar)^3} \int d^3p e^{i\mathbf{p} \cdot \mathbf{r}}$
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 $U(r) = \frac{1}{(2\pi\hbar)^3} \int d^3p e^{i\mathbf{p} \cdot \mathbf{r}}$

1d) action q_i

System made of n identical particles

$H(r_1, r_2, \dots, r_n)$
 the action is given by
 $S = \int_{t_1}^{t_2} L dt = \int_{t_1}^{t_2} (T - V) dt$

H is the Hamiltonian operator. It acts on the state ψ and gives the energy E .
 $H \psi = E \psi$
 $H = \frac{1}{2m} \nabla^2 + V(r)$
 $H = \frac{1}{2m} \nabla^2 + V(r)$
 where $V(r)$ is the potential energy.

$H = \frac{1}{2m} \nabla^2 + V(r)$
 $H \psi = E \psi$
 $H = \frac{1}{2m} \nabla^2 + V(r)$
 $H \psi = E \psi$

Constructing Hamiltonian for a system of n particles

$H = \frac{1}{2m} \nabla^2 + V(r)$
 $H \psi = E \psi$

$H = \frac{1}{2m} \nabla^2 + V(r)$
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$H = \frac{1}{2m} \nabla^2 + V(r)$
 $H \psi = E \psi$

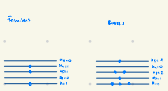
What happens if the particles are in the same position? (they are not distinguishable)

First case: $H = \frac{1}{2m} \nabla^2 + V(r)$ for two particles. The Hamiltonian for two particles is the sum of the Hamiltonians for each particle. The wave function is the product of the wave functions for each particle.

Second case: $H = \frac{1}{2m} \nabla^2 + V(r)$ for two particles. The Hamiltonian for two particles is the sum of the Hamiltonians for each particle. The wave function is the product of the wave functions for each particle.

Examples:
 $H = \frac{1}{2m} \nabla^2 + V(r)$
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$z = (2\pi)^{-3} \int d^3p e^{i\mathbf{p} \cdot \mathbf{r}} = \delta^3(\mathbf{r})$
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Hamiltonian operator

$H = \frac{1}{2m} \nabla^2 + V(r)$
 $H = \frac{1}{2m} \nabla^2 + V(r)$

Examples

$H = \frac{1}{2m} \nabla^2 + V(r)$
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1d) with particles

$H = \frac{1}{2m} \nabla^2 + V(r)$
 $H = \frac{1}{2m} \nabla^2 + V(r)$

where $G = \frac{1}{2} \int_{-\infty}^{\infty} dx \psi^2(x)$

Geometrisch anschauen: $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \dots$
 $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \dots$
 $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \dots$

Rechnerisch anschauen: $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \dots$
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